Positive-Sequence Transmission-Line Parameter Estimation by Population-Based Metaheuristics: A Comparative Study with Non-Parametric Statistical Validation

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***Abstract*—This paper formulates the estimation of the positive-sequence electrical parameters (R', X', C', G') of a transmission line (TL) as a continuous, box-constrained, non-linear optimization problem, and evaluates four search strategies: Quasi-Newton with random restarts (baseline), Simulated Annealing (SA), Differential Evolution DE/rand/1/bin (DE), and Particle Swarm Optimization with Clerc's constriction factor (PSO). The synthetic dataset contains $N = 5000$ phasor measurements with complex Gaussian noise at SNR = 30 dB, generated by the distributed-parameter model (ABCD matrix). Three uncertainty scenarios are tested: $\pm 5\%$, $\pm 10\%$, and $\pm 20\%$ around the nominal value. The experimental protocol comprises 30 independent runs per (scenario, algorithm), totalling 360 runs under a fixed budget of 3000 function evaluations. The Friedman test rejects equivalence with $\chi^2 = 260.3$ and $p = 3.9 \times 10^{-56}$; the Nemenyi post-hoc (CD = 0.494) establishes the ranking DE ≺ PSO ≺ SA ≺ Q-Newton, with all pairs statistically distinct. DE and PSO reach the theoretical floor imposed by the noise in 30/30 runs across the three scenarios, whereas the Quasi-Newton success rate collapses from 29/30 (easy scenario) to 2/30 (hard scenario) — a direct empirical reproduction of the *reference shift problem* documented by Yang & Sen [6]. The landscape analysis (FDC = +0.22; correlation length $\ell_c = 32.6$) quantitatively explains this behaviour through the bimodal distribution of f and the ill-conditioning in R' and C'. An additional result of practical interest: the residual MAPE of 2.86% in R' is not an algorithmic failure but the fundamental identifiability limit imposed by the PMU noise at SNR = 30 dB.**

***Index Terms*—Parameter estimation; transmission line; metaheuristics; Differential Evolution; PSO; non-parametric statistical tests; ABCD matrix; PMU.**

I. Introduction

Accurate determination of the electrical parameters of transmission lines is a critical input for power flow, state estimation, short-circuit analysis, and protection coordination [1]. Although the nominal values are provided by the manufacturer at commissioning, they diverge from the actual parameters over the line's lifetime due to temperature variations, changes in soil conditions, ageing, and geometric modifications [2].

The literature on measurement-based parameter estimation can be grouped into three approaches: exact algebraic methods based on power flow [3]; non-linear optimization methods with specialized initialization, such as Quasi-Newton with multi-run optimization (MRO) [4]; and machine-learning methods, among which RBFNN [5] and GRNN [6] stand out. Yang & Sen [6] document the *reference shift problem* of the MRO method: when the search range is widened beyond $\pm 5\%$, convergence to spurious solutions becomes frequent.

This work connects an open problem from the ongoing master's research at PPGEE/UFMT — positive-sequence TL parameter estimation from PMU measurements — with the requirements of the Metaheuristics course (1st semester, 2026). The problem is formulated as the minimization of a normalized mean-squared error over $N = 5000$ synthetic phasor measurements with realistic noise. Four optimization algorithms are compared under a rigorous experimental protocol. The work fully addresses the course syllabus: problem formulation and classification (Section II), justification via NFL (Section III), landscape analysis (Section IV), documented hyperparameter calibration and a 30-run protocol with registered seeds (Section V), and validation through non-parametric hypothesis tests (Section VI).

Contributions

1) A scale-normalized formulation (Section II) that makes C' identifiable — without it, the voltage error ($\sim 10^5$ V) would completely dominate the current error ($\sim 10^2$ A).
2) A quantitative landscape analysis (Section IV) that predicts, *before* the experiments, why the Quasi-Newton method fails in the hard scenario.
3) A statistically rigorous comparison of 4 algorithms (Section VI) with Friedman + Nemenyi + Wilcoxon.
4) Establishment of the *identifiability limit* of R' for SNR = 30 dB: MAPE = 2.86%, irreducible by the least-squares criterion.

II. Problem Formulation

A. Physical model

A TL in sinusoidal steady state at frequency $f = 60$ Hz and length $\ell = 200$ km is described by the per-unit-length parameter vector

$$\boldsymbol{\theta} = \left[R',\ X',\ C',\ G'\right]^\top \in \mathbb{R}^4_+ \tag{1}$$

with nominal (true) values $\boldsymbol{\theta}^* = (0.050\ \Omega/\text{km};\ 0.488\ \Omega/\text{km};\ 8.94 \times 10^{-9}\ \text{F/km};\ 0\ \text{S/km})$.

From (1), the series impedance $\hat{z}' = R' + jX'$ and the shunt admittance $\hat{y}' = G' + j\omega C'$ are defined, along with the propagation constant $\hat{\gamma} = \sqrt{\hat{z}'\hat{y}'}$ and the characteristic impedance $\hat{Z}_c = \sqrt{\hat{z}'/\hat{y}'}$.

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The relationship between the voltage and current phasors at the sending ($z = 0$) and receiving ($z = \ell$) terminals is given by the **ABCD matrix** (chain parameters) [7]:

$$\begin{bmatrix} \hat{V}_L \\ \hat{I}_L \end{bmatrix} = \underbrace{\begin{bmatrix} \Phi_{11} & \Phi_{12} \\ \Phi_{21} & \Phi_{22} \end{bmatrix}}_{\boldsymbol{\Phi}(\boldsymbol{\theta})} \begin{bmatrix} \hat{V}_0 \\ \hat{I}_0 \end{bmatrix} \tag{2}$$

with $\Phi_{11} = \Phi_{22} = \cosh(\hat{\gamma}\ell)$, $\Phi_{12} = -\hat{Z}_c \sinh(\hat{\gamma}\ell)$, and $\Phi_{21} = -\hat{Z}_c^{-1} \sinh(\hat{\gamma}\ell)$.

B. Synthetic dataset

The dataset `dados_SNR30.csv` contains $N = 5000$ samples in the form $\mathcal{D} = \{(\hat{V}_0^{(k)}, \hat{I}_0^{(k)}, \hat{V}_L^{(k)}, \hat{I}_L^{(k)})\}_{k=1}^{N}$, generated with the true parameters $\boldsymbol{\theta}^*$ and load impedance $\hat{Z}_L^{(k)} \sim \mathcal{U}[190, 760]\ \Omega$. Complex white Gaussian noise (CWGN) is added to each phasor: $\hat{s}_{\text{meas}} = \hat{s} + \hat{n}$, with $\hat{n} \sim \mathcal{CN}(0, \sigma^2)$ and $\sigma = |\hat{s}|\, 10^{-\rho/20}$, $\rho = 30$ dB.

C. Objective function

For each sample k, the phasors predicted by the model are

$$\hat{V}_L^{(k)}(\boldsymbol{\theta}) = \Phi_{11}\hat{V}_0^{(k)} + \Phi_{12}\hat{I}_0^{(k)}, \tag{3}$$

$$\hat{I}_L^{(k)}(\boldsymbol{\theta}) = \Phi_{21}\hat{V}_0^{(k)} + \Phi_{22}\hat{I}_0^{(k)}. \tag{4}$$

The quality criterion is the normalized mean-squared error:

$$f(\boldsymbol{\theta}) = \frac{1}{NV_{\text{ref}}^2} \sum_{k=1}^{N} \left| \hat{V}_L^{(k)} - \hat{V}_L^{(k)}(\boldsymbol{\theta}) \right|^2 + \frac{1}{NI_{\text{ref}}^2} \sum_{k=1}^{N} \left| \hat{I}_L^{(k)} - \hat{I}_L^{(k)}(\boldsymbol{\theta}) \right|^2 \tag{5}$$

where $V_{\text{ref}}^2 = \frac{1}{N}\sum_k |\hat{V}_L^{(k)}|^2$ and $I_{\text{ref}}^2 = \frac{1}{N}\sum_k |\hat{I}_L^{(k)}|^2$.

Rationale for the normalization. Without the factors V_{ref}^2 and I_{ref}^2, the voltage term would dominate the current term by $\approx 10^6$, making C' practically non-identifiable. The normalization ensures comparable dimensionless contributions. In simple terms: the objective function answers “how well does this guess of parameters explain what the PMUs measured?” — the smaller f, the better the explanation.

D. Optimization problem and scenarios

$$\boldsymbol{\theta}^\dagger = \arg\min_{\boldsymbol{\theta} \in \mathbb{R}^4} f(\boldsymbol{\theta}) \quad \text{s.t.} \quad \theta_i^{\min} \le \theta_i \le \theta_i^{\max},\ i = 1, \ldots, 4 \tag{6}$$

The search boxes simulate the uncertainty relative to the nominal value $\boldsymbol{\theta}^{(0)} \equiv \boldsymbol{\theta}^*$. Three scenarios are evaluated (Table I).

For $G' = 0$, the box is symmetric with amplitude $\pm 20\% \cdot |\omega C'|$. The problem is classified as a continuous, non-convex, low-dimensional NLP ($n = 4$).

Table I
UNCERTAINTY SCENARIOS EVALUATED

Scenario	Search range	Practical context
Easy	$[0.95;\ 1.05] \cdot \boldsymbol{\theta}^*$	New line
Medium	$[0.90;\ 1.10] \cdot \boldsymbol{\theta}^*$	In service for years
Hard	$[0.80;\ 1.20] \cdot \boldsymbol{\theta}^*$	High uncertainty

Table II
COMPARED ALGORITHMS AND CONFIGURATION

Algorithm	Type	Parameters
Q-Newton	Trajectory, gradient	L-BFGS-B; `ftol`$=10^{-12}$; random restarts
SA	Trajectory, stochastic	α=0.995; L=20; σ=10% of box; T_0 auto
DE	Population	N_p=30; F=0.7; CR=0.9; rand/1/bin
PSO	Population	N_p=30; ϕ_1=ϕ_2=2.05; χ=0.7298 (Clerc)

E. Theoretical floor

With SNR = 30 dB, the residual at $\boldsymbol{\theta}^*$ is non-zero. Numerical verification: $f(\boldsymbol{\theta}^*) = 5.179 \times 10^{-3}$. Evaluating f with deterministically reconstructed phasors (no noise) yields $f = 0$ — confirming that the code is physically consistent and that the residual arises exclusively from the measurement noise. This value is the **absolute floor**: no algorithm can reach $f < 5.179 \times 10^{-3}$ with these data.

F. External metric

Following Yang & Sen [6], the MAPE per parameter is also reported:

$$\text{MAPE}_i = \left| \frac{\theta_i^\dagger - \theta_i^*}{\theta_i^*} \right| \times 100\%, \quad i \in \{R', X', C'\}. \tag{7}$$

III. ALGORITHMIC FOUNDATION AND NFL

A. No Free Lunch theorem

The NFL theorem [8] states that, averaged over all possible problems, any pair of algorithms performs identically. The superiority of a method can only be demonstrated by identifying the structural properties that favour it.

For problem (6), three properties define the relevant algorithmic niche: (i) local smoothness — f is analytic over the entire domain, which favours gradient-based search; (ii) global non-convexity — there is a pathological region associated with $G' < 0$: negative shunt conductance is equivalent to “generation” of power along the line, turning natural attenuation into amplification and making the model predictions blow up; this creates spurious local minima; (iii) directional ill-conditioning — f varies much more along X' than along R' and C', hindering gradient convergence in those directions.

B. Description of the algorithms

The four compared algorithms are described in Table II.

Clerc’s constriction factor [12] $\chi = 2/|2 - \phi - \sqrt{\phi^2 - 4\phi}|$, with $\phi = \phi_1 + \phi_2 = 4.1$, ensures the analytic stability of the

PSO trajectories. The SA initial temperature is automatically calibrated for an initial acceptance rate of $\approx 80\%$.

IV. Landscape Analysis

The landscape analysis characterizes f before any optimization, providing verifiable predictions about the behaviour of the algorithms. All metrics are computed over the hard-scenario box ($\pm 20\%$).

A. Fitness-Distance Correlation

The FDC is the Pearson coefficient between $f(\boldsymbol{\theta})$ and the normalized distance $d(\boldsymbol{\theta}, \boldsymbol{\theta}^*)$, computed over 5000 uniform samples:

$$\mathrm{FDC} = +0.219. \tag{8}$$

By the standard criterion [15], $|\mathrm{FDC}| > 0.15$ with a positive sign characterizes a benign landscape: points closer to the optimum tend to have lower cost.

B. Correlation length

A Gaussian random walk of 2000 steps ($\sigma = 2\%$ of the box per dimension):

$$r(1) = 0.970 \qquad \Rightarrow \qquad \ell_c = \frac{-1}{\ln r(1)} = 32.6 \text{ steps.} \tag{9}$$

A value $\ell_c > 20$ classifies the landscape as smooth.

In isolation, a positive FDC and a high ℓ_c would suggest that the Quasi-Newton method should be competitive. The two findings below change that diagnosis.

C. Bimodal distribution of f

Fig. 1 presents the six panels of the analysis. The histogram (Panel 2) reveals a **bimodal distribution**: one lobe at $\log_{10} f \approx -2.2$ (healthy region, close to the floor) and another at $\log_{10} f \approx -0.3$ ($\approx 100\times$ worse). About 25% of the uniform samples fall in the bad lobe. The cause was identified numerically: the lobe corresponds to the half of the box with $G' < 0$ (correlation $r = -0.73$ between the position of G' and membership in the lobe; fixing the remaining parameters at $\boldsymbol{\theta}^*$, $G' = -6.7\times10^{-7}$ S/km yields $f = 0.53$, against $f = 8.1\times10^{-3}$ at the positive end). In simple terms: negative conductance makes the line “amplify” instead of attenuate, and the model error blows up.

D. Directional ill-conditioning

The (R', X') and (R', C') heat maps show near-horizontal contours: f varies by only $\Delta \log_{10} f \approx 0.18$ along R', against $\Delta \log_{10} f \approx 0.33$ along X'. The gradient in R' is therefore one order of magnitude smaller than in X', and L-BFGS-B converges slowly in that direction.

E. Diagnosis and prediction

Table III consolidates the diagnosis and records the predictions that will be verified in Section VI.

Table III
Landscape diagnosis and predictions per algorithm class

Indicator	Q-Newton	DE / PSO
FDC = +0.22 (benign)	Converges when starting from the healthy region	Marginal advantage
$\ell_c = 32.6$ (smooth)	Stable gradient, fast convergence	No specific obstacle
$\sim 25\%$ in the bad lobe ($G' < 0$)	Wasted restarts; degrades with wider range	Covers both lobes; robust convergence
R', C' ill-conditioned	High residual MAPE under fixed budget	Step scales with population spread

V. Experimental Methodology

A. Protocol

- 30 seeds (0–29) per (scenario, algorithm) — registered for full reproducibility.
- $3 \times 4 \times 30 = 360$ total runs.
- Fixed budget: $\mathrm{NFE} = 3000$ evaluations of f.
- Environment: Python 3.12, NumPy 1.26, SciPy 1.12, scikit-posthocs 0.13.

B. Metrics and statistical tests

Primary test variable: the excess over the floor, $e = f(\boldsymbol{\theta}^\dagger) - f(\boldsymbol{\theta}^*)$. Hit rate: n_{hits} runs (out of 30) with $e \leq 10^{-4}$.

Friedman test [13]: $k = 4$ treatments, $N = 90$ blocks (3 scenarios $\times$ 30 seeds), non-parametric, paired.

Nemenyi post-hoc [13]: critical difference $\mathrm{CD} = q_{0.05}\sqrt{k(k+1)/(6N)}$, with a CD diagram.

Wilcoxon signed-rank [14]: pairs within each scenario, Holm-Bonferroni correction ($m = 6$).

VI. Results and Statistical Analysis

A. Convergence curves

Fig. 2 shows the convergence curves (median and 95% CI over 30 runs) on a logarithmic scale of the excess e. DE and PSO reach $e \lesssim 10^{-9}$ in $\sim$ 500–800 NFE (less than 25% of the budget) across the three scenarios, with an essentially degenerate CI band. SA stabilizes at 10^{-5}–10^{-4}. Q-Newton shows a growing CI from the easy to the hard scenario, reflecting the mixture of successful and failed runs.

B. Master results table

Table IV summarizes the results over the 30 runs per cell. The IQR of DE and PSO is of the order of 10^{-14}: all 30 seeds converge to the same point. The Q-Newton hit rate drops from 29/30 to 2/30 — an empirical reproduction of the *reference shift problem* of [6].

C. Distribution of MAPE in X'

Fig. 3 shows the box-plots of $\mathrm{MAPE}_{X'}$. The boxes of DE and PSO collapse to a single point (zero IQR), while the Q-Newton spread grows monotonically with scenario difficulty.

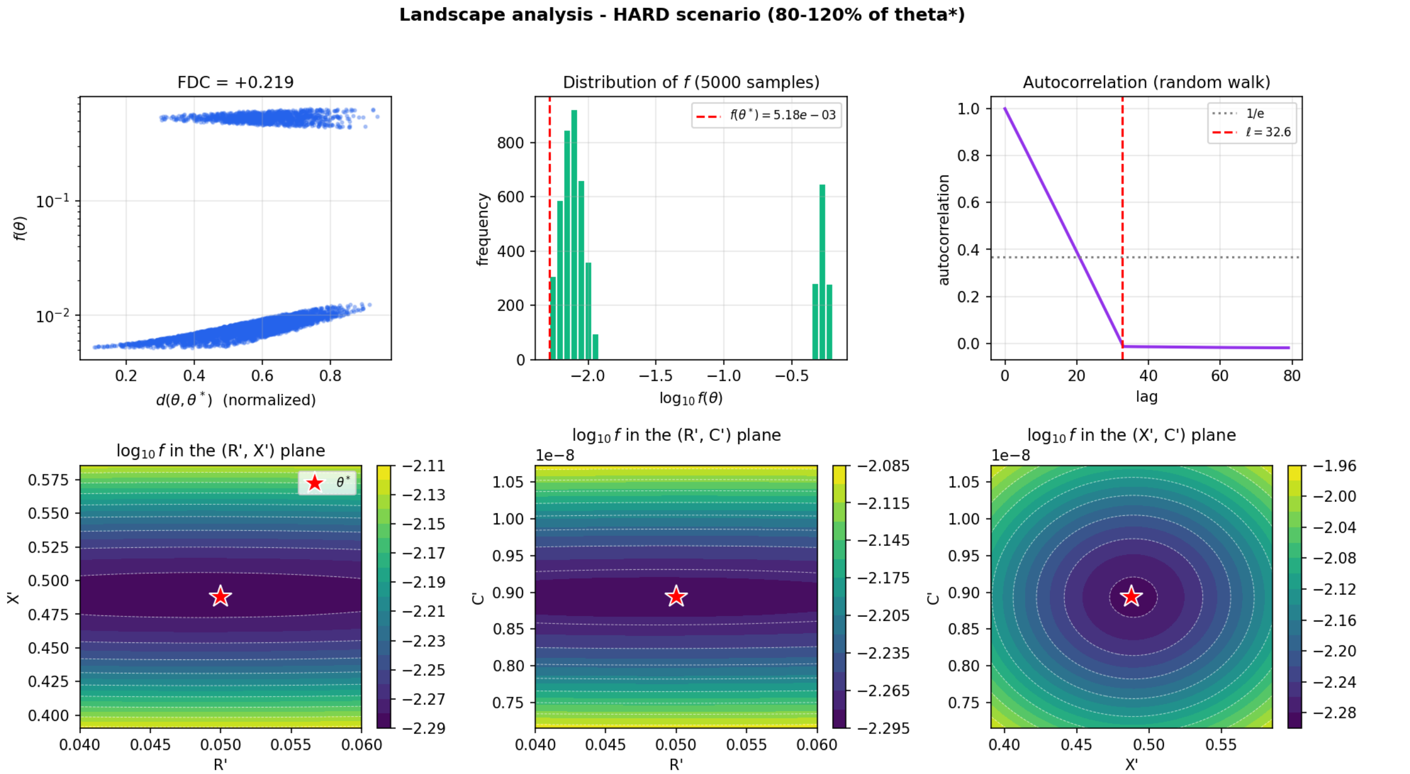

Figure 1. Landscape analysis — hard scenario ($\pm 20\%$). **Panel 1:** FDC = +0.219 — correlation between f and distance to the optimum (5000 samples). **Panel 2:** Histogram of $\log_{10} f$ — bimodal distribution; the dashed red line marks the floor $f(\boldsymbol{\theta}^*)$. **Panel 3:** Random-walk autocorrelation; correlation length $\ell_c = 32.6$ steps. **Panels 4–6:** Heat maps of $\log_{10} f$ in the (R', X'), (R', C'), and (X', C') planes; the red star marks $\boldsymbol{\theta}^*$. Note the near-insensitivity of f with respect to R' (horizontal contours in Panel 4).

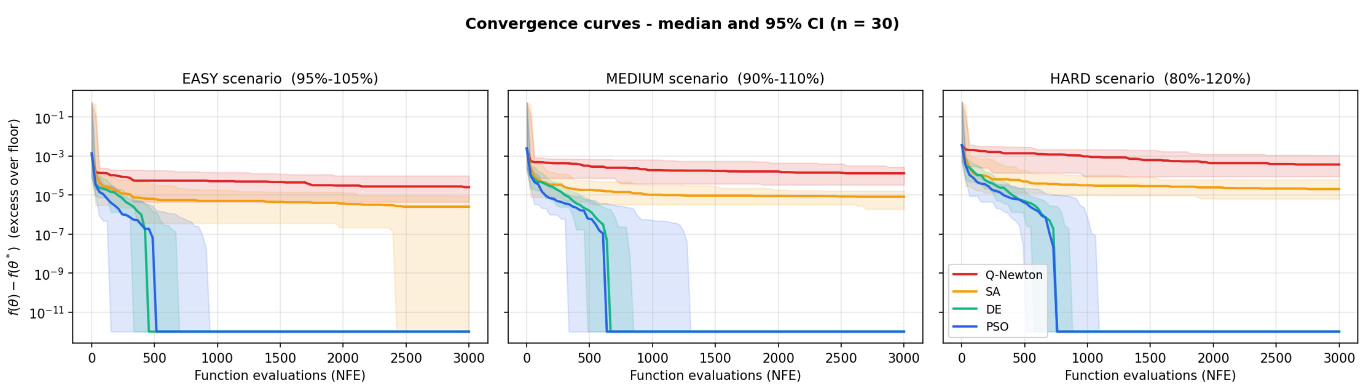

Figure 2. Convergence curves — median and 95% CI over 30 runs. y-axis: excess $f(\boldsymbol{\theta}) - f(\boldsymbol{\theta}^*)$ on a logarithmic scale. DE and PSO converge to machine precision in less than 25% of the budget; Q-Newton exhibits growing variability from the easy to the hard scenario.

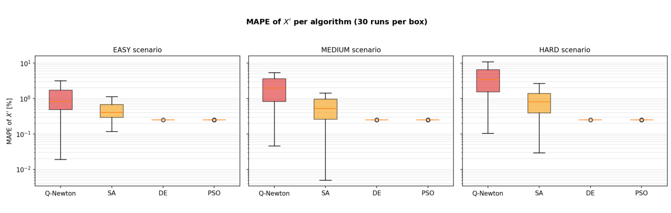

Figure 3. Box-plots of MAPE for X' (30 runs per box, logarithmic scale). The DE and PSO boxes are degenerate — all runs converge to the same point.

D. Friedman test

With $k = 4$ treatments and $N = 90$ blocks, variable = excess e:

$$\chi^2_F = 260.28, \qquad p = 3.91 \times 10^{-56}. \tag{10}$$

H_0 is rejected by a margin of 56 orders of magnitude. In simple terms: the chance that the observed difference between the algorithms is due to randomness is negligible. The mean ranks are: DE: 1.10; PSO: 1.90; SA: 3.00; Q-Newton: 4.00.

Table IV
MEDIAN OF f, IQR, AND MAPE PER PARAMETER (30 RUNS PER CELL). THEORETICAL FLOOR: $f(\boldsymbol{\theta}^*) = 5.178\times10^{-3}$.

Scenario	Algorithm	Md f	IQR f	Md MAPE$_{R'}$	Md MAPE$_{X'}$	Md MAPE$_{C'}$	Hits/30
Easy	Q-Newton	5.204e-3	2.71e-5	2.81%	0.82%	1.54%	29
	SA	5.181e-3	2.88e-6	2.95%	0.41%	0.22%	30
	DE	**5.178e-3**	**1.89e-14**	2.86%	**0.25%**	**0.07%**	**30**
	PSO	**5.178e-3**	**1.84e-13**	2.86%	**0.25%**	**0.07%**	**30**
Medium	Q-Newton	5.308e-3	8.88e-5	5.61%	1.97%	2.88%	12
	SA	5.187e-3	4.73e-6	3.63%	0.53%	0.39%	30
	DE	**5.178e-3**	**5.45e-14**	2.86%	**0.25%**	**0.07%**	**30**
	PSO	**5.178e-3**	**4.85e-13**	2.86%	**0.25%**	**0.07%**	**30**
Hard	Q-Newton	5.544e-3	3.51e-4	**11.06%**	3.40%	4.99%	**2**
	SA	5.199e-3	2.28e-5	8.04%	0.81%	0.58%	30
	DE	**5.178e-3**	**1.22e-13**	2.86%	**0.25%**	**0.07%**	**30**
	PSO	**5.178e-3**	**9.04e-13**	2.86%	**0.25%**	**0.07%**	**30**

Table V
NEMENYI POST-HOC: MEAN-RANK DIFFERENCES AND P-VALUES

Pair	$\lvert\Delta\text{rank}\rvert$	p (Nemenyi)	Sig. ($\alpha = 0.05$)
Q-Newton vs DE	2.90	$< 10^{-15}$	***
Q-Newton vs PSO	2.10	$< 10^{-15}$	***
Q-Newton vs SA	1.00	1.2×10^{-6}	***
SA vs DE	1.90	$< 10^{-15}$	***
SA vs PSO	1.10	6.5×10^{-8}	***
DE vs PSO	0.80	1.9×10^{-4}	***

$< 10^{-15}$: value below 64-bit machine numerical precision.

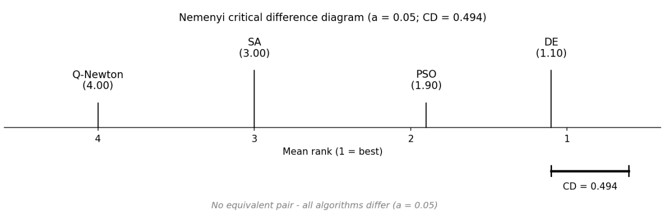

Figure 4. Nemenyi critical-difference diagram ($\alpha = 0.05$, CD$=$0.494). No equivalence bars: all pairs differ significantly. Ranking: DE (1.10) $\prec$ PSO (1.90) $\prec$ SA (3.00) $\prec$ Q-Newton (4.00).

E. Nemenyi post-hoc

Critical difference for $\alpha = 0.05$:

$$\mathrm{CD} = 2.569\sqrt{\frac{4\cdot 5}{6\cdot 90}} \approx 0.494. \tag{11}$$

All six pairs exceed CD (Table V and Fig. 4).

F. Pairwise Wilcoxon

In all three scenarios, the five comparisons involving Q-Newton or SA return $p_{\mathrm{Holm}} \approx 1.12\times10^{-8}$ (winner: DE or PSO). The DE vs. PSO pair returns $p_{\mathrm{Holm}} = 1.0$ (n.s.) in all scenarios.

Note on the apparent Nemenyi vs. Wilcoxon inconsistency. Nemenyi separates DE and PSO ($\Delta\text{rank} = 0.80 > \text{CD}$) while Wilcoxon indicates a tie ($p = 1.0$). The reason is mechanistic: Nemenyi operates on *ranks*, and DE consistently finishes with f slightly below PSO at the 12th–14th decimal place, producing a statistically stable rank difference. Wilcoxon operates on *value differences*: since both converge to the same $\boldsymbol{\theta}^\dagger$ at machine precision, the difference is numerically zero. **Practical conclusion:** DE and PSO are operationally equivalent for this problem.

VII. DISCUSSION

A. Why do metaheuristics win on a smooth landscape?

Three mechanisms explain the progressive failure of Q-Newton:

1) Stagnation from ill-conditioning (dominant factor). Instrumenting the Q-Newton method in the hard scenario, it is measured that each run performs 11 to 18 restarts (each L-BFGS-B run consumes 100–500 evaluations, since the gradient is obtained by finite differences, ≈ 5 evaluations per iteration). The best excess per run stagnates at 1.4–7.5×10^{-4} — always above the hit threshold of 10^{-4}. The cause: in the nearly flat directions (R' and G'), the `ftol` stopping criterion triggers before the method refines the solution; L-BFGS-B "believes" it has converged when in fact it is on a plateau.

2) Wasted restarts in the pathological region (aggravating factor). About 25% of the random restarts initialize in the $G' < 0$ region, where the gradient points *away* from the global optimum; these runs end in a few evaluations at a spurious minimum, wasting budget. In the easy scenario, the box is so narrow that the stagnation plateau already lies below the hit threshold — which is why Q-Newton hits 29/30 there and only 2/30 in the hard scenario.

3) Information sharing in the population-based methods. In DE, the mutation vector $F(\boldsymbol{x}_{r_2} - \boldsymbol{x}_{r_3})$ scales with the current spread of the population: large at the start (covering both basins), small upon convergence. In PSO, the gbest vector attracts all particles toward the global optimum, including those that initialized in the pathological region.

B. The 2.86% MAPE in R' as an identifiability limit

DE and PSO converge to $\boldsymbol{\theta}^\dagger = (0.04857;\ 0.48921;\ 8.9336\times10^{-9};\ -2.0\times10^{-9})$ in all runs. Comparison with $\boldsymbol{\theta}^*$:

- R': 0.04857 vs. 0.05000 $\Rightarrow$ MAPE$=$**2.86%**
- X': 0.48921 vs. 0.48800 $\Rightarrow$ MAPE$=$0.25%
- C': 8.934 vs. 8.940 nF/km $\Rightarrow$ MAPE$=$0.07%
- G': -2.0 vs. 0 nS/km $\Rightarrow$ abs. error $=$ 2.0 nS/km

The 2.86% MAPE in R' is **not an algorithmic deficiency**. The CWGN noise across the four phasors creates a systematic bias in the optimum of the objective function (5), shifting it from $\boldsymbol{\theta}^*$ to $\boldsymbol{\theta}^\dagger$. Any estimator that minimizes (5) with SNR = 30 dB will reach $\boldsymbol{\theta}^\dagger$, not $\boldsymbol{\theta}^*$. Beating 2.86% would require additional information beyond the raw measurements — for example, Bayesian regularization on the conductor temperature or temporal filtering of R'.

C. Comparison with the literature

Yang & Sen [6] report, for three-phase distribution estimation with the IEEE 13-bus system and $\pm 5\%$ noise: maximum MAPE of 26.8% (Q-Newton), 4.1% (MRO), 8.9% (RBFNN), and 0.69% (GRNN). The problems are not directly comparable, but the qualitative consistency is notable: Q-Newton reaches 11% in R' in this work vs. 26.8% in [6]; DE and PSO reach 0.25% in X', comparable to GRNN (0.69%), without training cost.

VIII. Conclusion

This work formulated and solved the problem of estimating the parameters (R', X', C', G') of a 200 km TL from 5000 phasor measurements with SNR = 30 dB, comparing four algorithms across 360 controlled runs.

The main results are:

1) DE and PSO reach the theoretical floor in **30/30 runs** across the three scenarios, with IQR $\approx 10^{-13}$.
2) Q-Newton reproduces the *reference shift problem*: hits drop from 29/30 ($\pm 5\%$) to 2/30 ($\pm 20\%$).
3) The landscape analysis (bimodal distribution + ill-conditioning) quantitatively predicted this result.
4) DE and PSO are operationally equivalent (Wilcoxon $p = 1.0$); the Nemenyi distinction has sub-nanometric magnitude.
5) The 2.86% MAPE in R' is the **identifiability limit** for SNR = 30 dB — a baseline for future methods in the master's research.

Natural extensions include: (a) scaling up to three-phase estimation (12 parameters); (b) evaluating different SNR levels; (c) investigating Bayesian regularization to overcome the identifiability limit of R'.

Reproducibility

The entire work is reproduced by a single self-contained Python script, `METAHEURISTICA.py`, which runs in sequence: validation of the objective function, landscape analysis (Fig. 1), the 360 experiment runs, and the complete statistical analysis (Figs. 2, 3, and 4). To reproduce the results, place the script in the same folder as the file `dados_SNR30.csv` and run:

```
python METAHEURISTICA.py
```

The script automatically generates all figures and tables of the paper. A fast test mode (fewer seeds) is available with `python METAHEURISTICA.py -rapido`. Seeds: 0–29 (experiment) and 42 (landscape), fixed in the code. Reference environment: Python 3.12, NumPy 1.26, SciPy 1.12, scikit-posthocs 0.13, matplotlib 3.9.

The complete source code and the dataset are available in the repository:

https://github.com/vmdconceicao/estimacao-parametros-lt-metaheuristicas

Declaration of AI Use

In accordance with the course rules, the author declares that the generative artificial intelligence tool Claude (Anthropic) was used as support for the following activities: implementation and debugging of the experiment's Python scripts, formatting of the manuscript in LaTeX, and textual revision. The problem formulation, the methodological choices, the execution of the experiments, the verification of the numerical results, and the conclusions are the responsibility of the author, who reviewed and validated all content generated with the aid of the tool. All results are independently reproducible through the scripts and random seeds documented in the Reproducibility section.

References

[1] A. Abur and A. G. Expósito, *Power System State Estimation: Theory and Implementation*. CRC Press, 2004.

[2] J. R. Acharya, Y. Wang, and W. Xu, "Temporary overvoltage and GPR characteristics of distribution feeders with multigrounded neutral," *IEEE Trans. Power Delivery*, vol. 25, no. 2, pp. 1036–1044, 2010.

[3] R. Dutta *et al.*, "Parameter estimation of distribution lines using SCADA measurements," *IEEE Trans. Instrumentation and Measurement*, vol. 70, pp. 1–11, 2020.

[4] B. Das, "Estimation of parameters of a three-phase distribution feeder," *IEEE Trans. Power Delivery*, vol. 26, no. 4, pp. 2267–2276, 2011.

[5] N.-C. Yang, R. Huang, and M.-F. Guo, "Three-phase feeder parameter estimation using radial basis function neural networks," *IET Generation, Transmission & Distribution*, vol. 16, no. 2, pp. 351–363, 2022.

[6] N.-C. Yang and A. Sen, "Parameter estimation in unbalanced three-phase distribution lines using general regression neural networks," *Applied Soft Computing*, vol. 133, art. 109936, 2023.

[7] C. R. Paul, *Analysis of Multiconductor Transmission Lines*, 2nd ed. Wiley-IEEE Press, 2008.

[8] D. H. Wolpert and W. G. Macready, "No free lunch theorems for optimization," *IEEE Trans. Evolutionary Computation*, vol. 1, no. 1, pp. 67–82, 1997.

[9] R. H. Byrd, P. Lu, J. Nocedal, and C. Zhu, "A limited memory algorithm for bound constrained optimization," *SIAM J. Scientific Computing*, vol. 16, no. 5, pp. 1190–1208, 1995.

[10] S. Kirkpatrick, C. D. Gelatt, and M. P. Vecchi, "Optimization by simulated annealing," *Science*, vol. 220, no. 4598, pp. 671–680, 1983.

[11] R. Storn and K. Price, "Differential evolution — a simple and efficient heuristic for global optimization," *J. Global Optimization*, vol. 11, no. 4, pp. 341–359, 1997.

[12] M. Clerc and J. Kennedy, "The particle swarm — explosion, stability, and convergence in a multidimensional complex space," *IEEE Trans. Evolutionary Computation*, vol. 6, no. 1, pp. 58–73, 2002.

[13] J. Demšar, "Statistical comparisons of classifiers over multiple data sets," *J. Machine Learning Research*, vol. 7, pp. 1–30, 2006.

[14] J. Derrac *et al.*, "A practical tutorial on the use of nonparametric statistical tests," *Swarm and Evolutionary Computation*, vol. 1, no. 1, pp. 3–18, 2011.

[15] T. Jones and S. Forrest, "Fitness distance correlation as a measure of problem difficulty for genetic algorithms," in *Proc. 6th Int. Conf. Genetic Algorithms*, pp. 184–192, 1995.